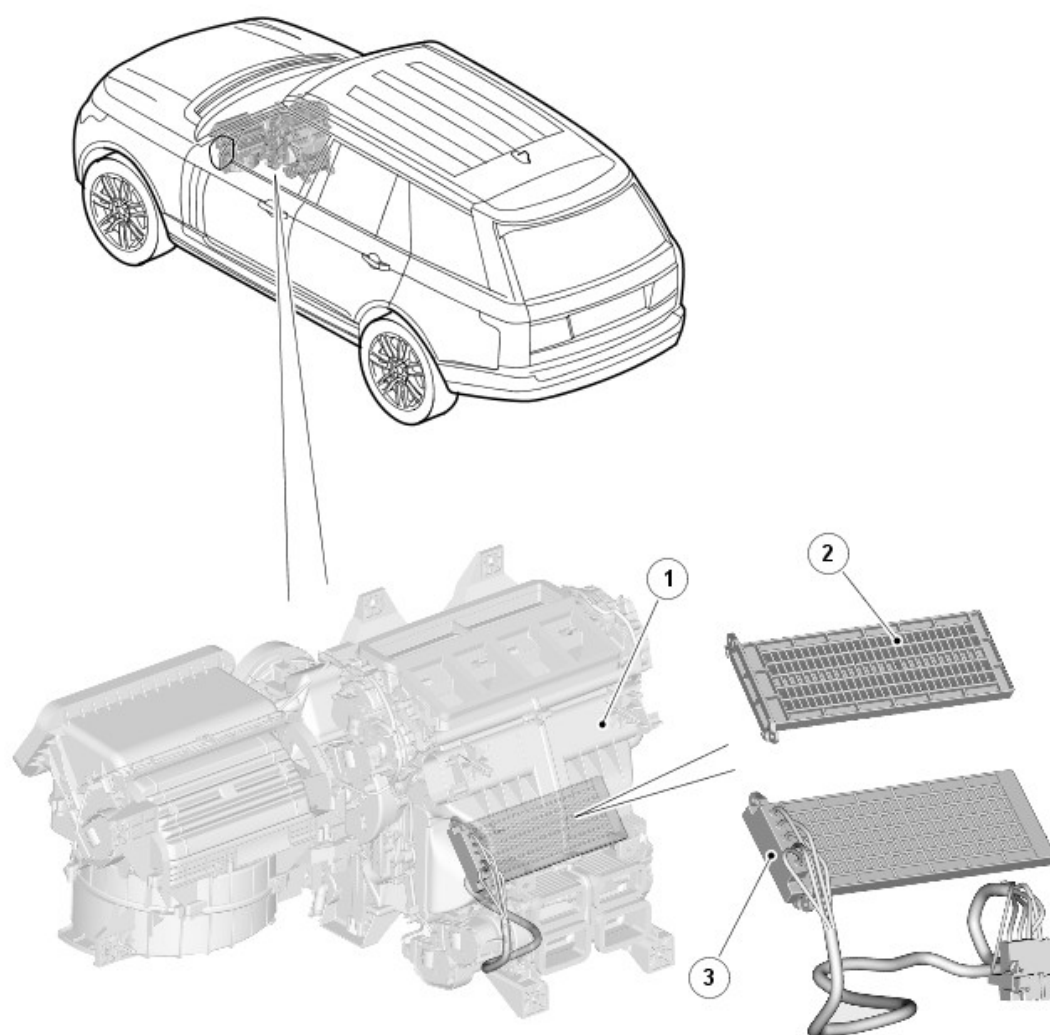


2016.0 RANGE ROVER (LG), 412-02

AUXILIARY CLIMATE CONTROL

DESCRIPTION AND OPERATION

COMPONENT LOCATION



E141419

ITEM	DESCRIPTION
1	Climate control assembly
2	Baffle plate
3	Electric booster heater

OVERVIEW

Depending on model specification and market, diesel models may incorporate auxiliary heating in the form of an electric booster heater. Where fitted, the electric booster heater boosts the temperature of the air in the climate control assembly, downstream of the heater core.

Electric booster heater operation is entirely automatic. Maximum heat output is approximately 10% of that available from the heater core when the engine is at normal operating temperature. If necessary, the output from the electric booster heater can be varied, to keep the load within what the battery and charging system can support.

DESCRIPTION

ELECTRIC BOOSTER HEATER

The electric booster heater consists of a matrix, containing four 150 W ceramic coated PTC (positive temperature coefficient) elements, which is installed in the left side of the climate control assembly. On vehicles without an electric booster heater, a baffle plate is installed in place of the booster heater.

The elements of the electric booster heater are wired together in pairs, with separate power feeds to each pair from two relays in the EJB (engine junction box). Operation of the relays is controlled by the ATCM (automatic temperature control module).

OPERATION

The ATCM (automatic temperature control module) uses the following inputs from the medium speed CAN (controller area network) comfort bus to control operation of the electric booster heater:

- Climate control system selections from the front and rear integrated control panels and the touch screen
- Ambient air temperature and ECT (engine coolant temperature) from the ECM (engine control module).
- Battery and generator status from the GWM (gateway module).
- Ignition mode and transit mode status from the CJB (central junction box).

The ATCM (automatic temperature control module) operates the electric booster heater, provided the vehicle is not in transit mode and there are no generator faults recorded, when the following conditions occur:

- The engine is running
- The blower is on (any speed)
- Heating is requested on the front or rear integrated control panel, or on the front or rear climate menu of the touch screen, and the desired vent temperature (calculated by the ATC (automatic temperature control) module), is not being achieved
- The ambient air temperature is less than 10 °C (50 °F)
- The ECT is less than 90 °C (194 °F)
- Battery charge state is more than 10 A.

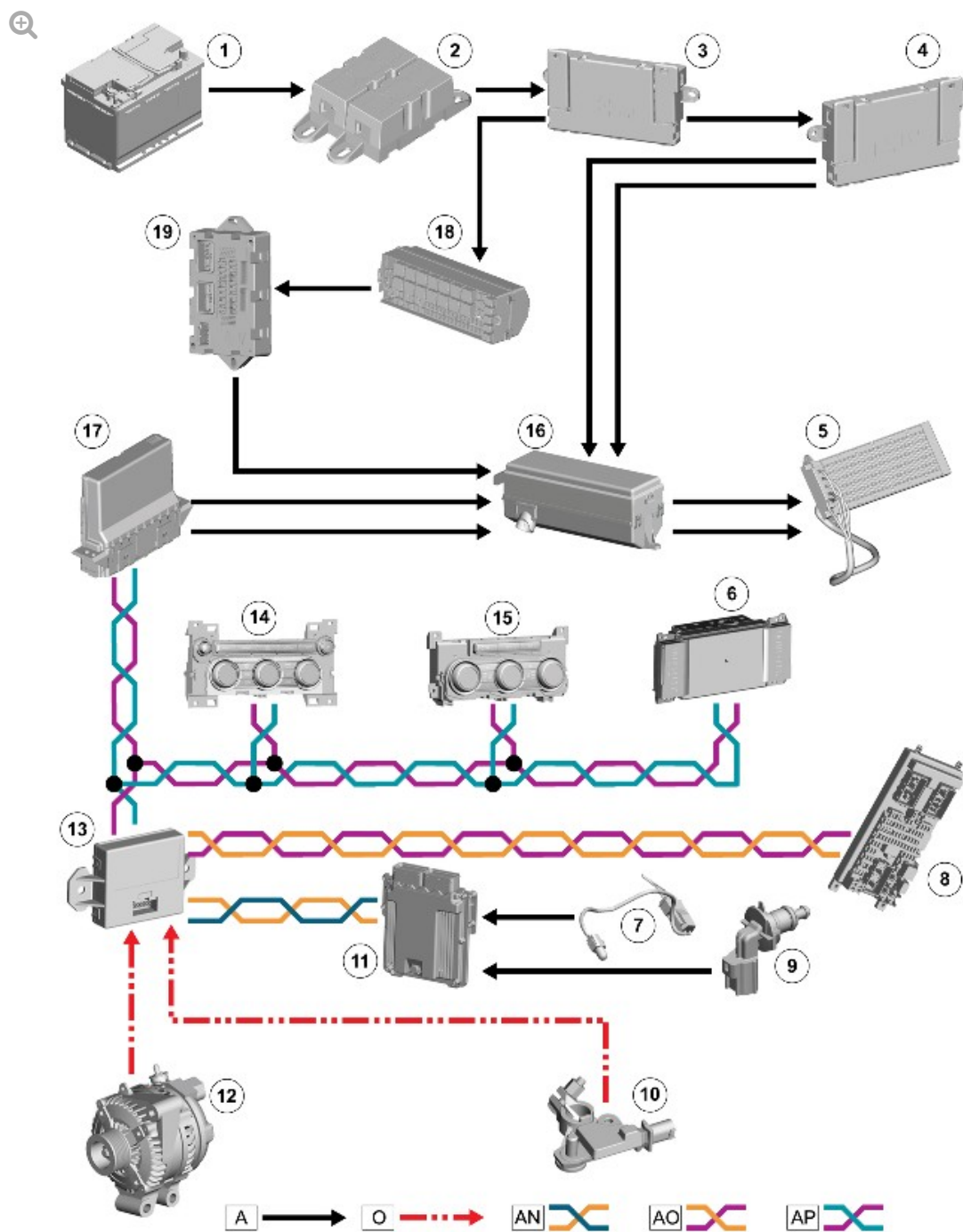
Depending on the battery charge state, the ATCM (automatic temperature control module) energizes one or both pairs of elements of the electric booster heater. If the battery charge state changes while the electric booster heating is in operation, the ATCM (automatic temperature control module) adjusts the number of elements being used accordingly, as detailed below:

BATTERY CHARGE STATE (A)	RELAY 1	RELAY 2
10 or less	OFF	OFF
More than 10 and up to 60	ON	OFF
More than 60	ON	ON

The ATCM (automatic temperature control module) discontinues operation of the electric booster heater if the ambient temperature increases to 15 °C (59 °F), or any of the other required conditions ceases to exist.

The ATCM (automatic temperature control module) monitors the control circuits of the electric booster heater relays and records a DTC (diagnostic trouble code) if it detects a short or open circuit. The load sides of the relays are not monitored, so any fault with these circuits or the electric booster heater will only be apparent by poor heating performance.

CONTROL DIAGRAM



A = HARDWIRED; O = LIN (LOCAL INTERCONNECT NETWORK) BUS; AN = HS (HIGH SPEED) CAN POWERTRAIN BUS; AO = MS (MEDIUM SPEED) CAN BODY BUS; AP = MS CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)

4	Auxiliary Junction Box (AJB)
5	Electric booster heater
6	Touch Screen (TS)
7	Ambient Air Temperature (AAT) sensor
8	Central Junction Box (CJB)
9	Engine Coolant Temperature (ECT) sensor
10	Battery Monitoring System (BMS) control module
11	Engine Control Module (ECM)
12	Generator
13	Gateway Module (GWM)
14	Integrated Control Panel (ICP)
15	Rear Integrated Control Panel (RICP)
16	Engine Junction Box (EJB)
17	Automatic Temperature Control Module (ATCM)
18	Rear Junction Box (RJB)
19	Quiescent Current Control Module (QCCM)